화학수학

# 세 번째 수업

## 다변수 미분

# 오늘의 수업

- Functions of several independent variables

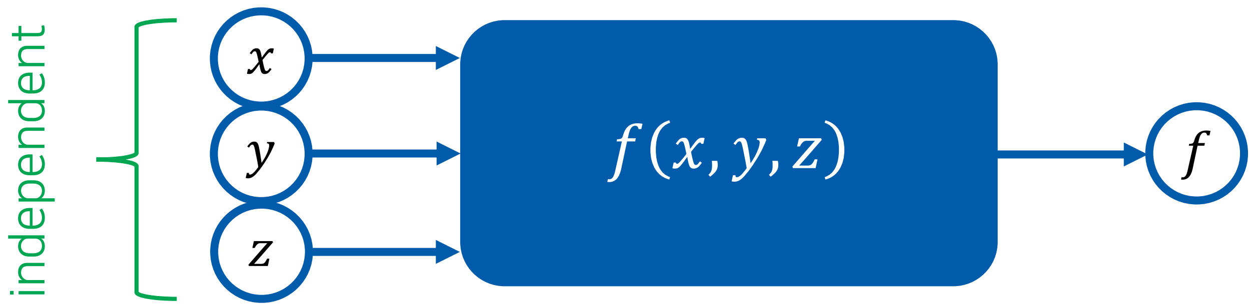
- Partial derivatives
  - Fundamental equation of differential calculus
  - Manipulating techniques

# 8. Differential Calculus with Several Independent Variables

(8장 다변수 미분)

# Function of Several Independent Variables

## (다변수함수)




Example: $f(x, y, z) = x^2 + xy + 3z$

$$f(1,1,1) = 1 + 1 + 3 = 5$$

$$f(-1,1,0) = 1 - 1 + 0 = 0$$

$$\vdots$$

# Examples from Physical Chemistry

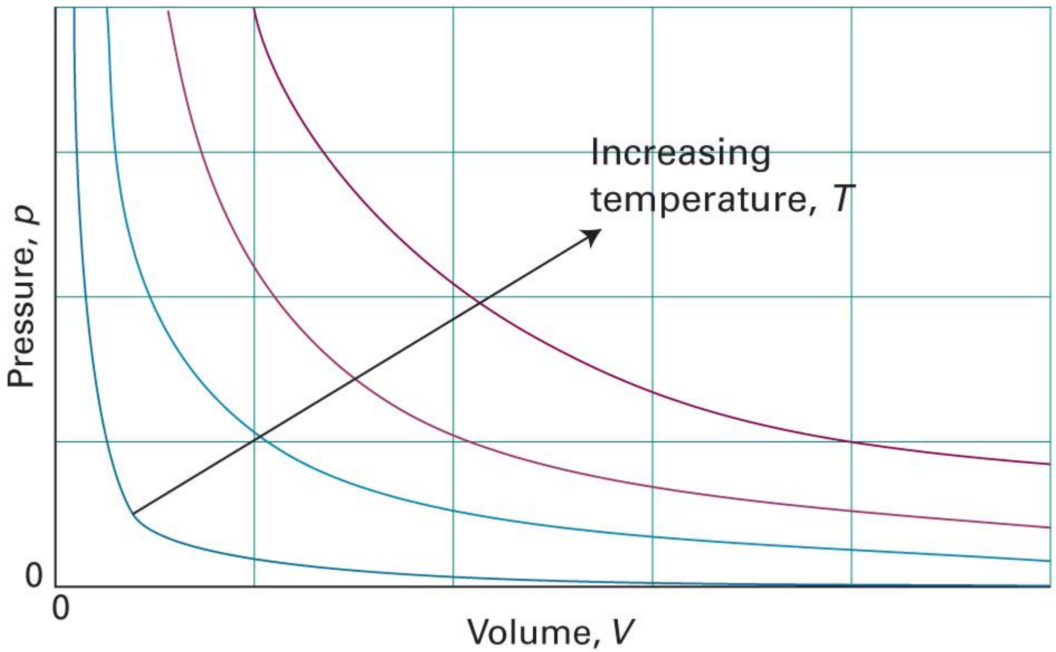
- Thermodynamic properties
  - Pressure $P = P(T, V, n)$
  - Internal energy $U = U(T, V, n)$
- Wave function $\psi = \psi(x, y, z)$
- Reaction rate $r = r([\mathrm{A}], [\mathrm{B}])$
- piecewise continuous and single-valued

# Example: Ideal Gas Equation

$$P = P(T, V, n) = \frac{nRT}{V}$$

# Visualization: Single Independent Variable

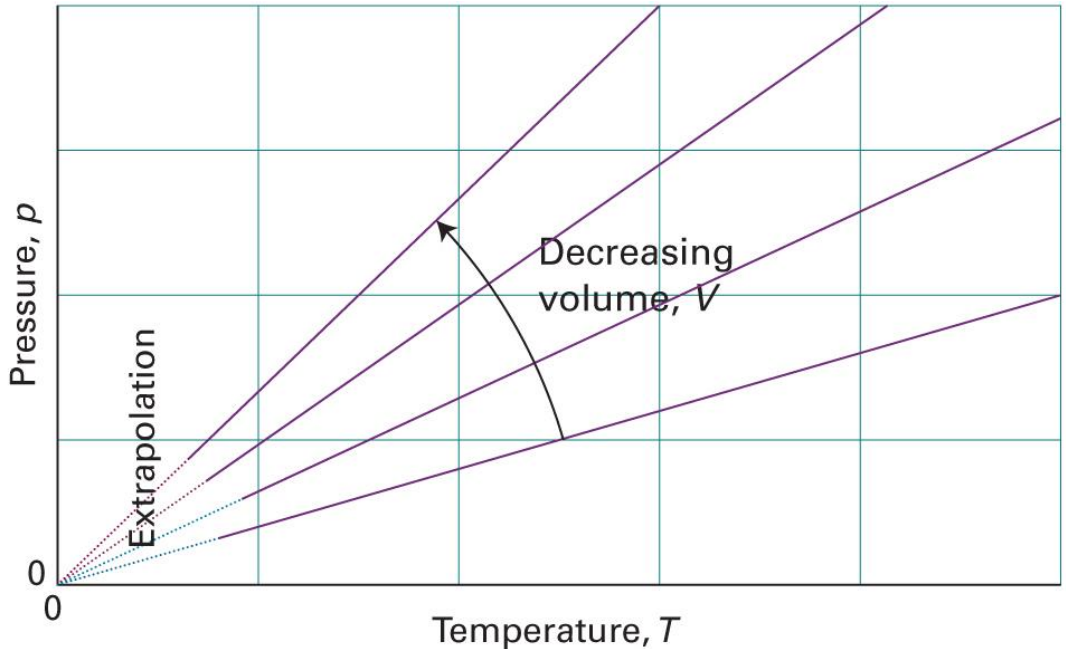
$$P = P(T, V, n) = \frac{nRT}{V}$$




(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 1A.2)

# Visualization: Single Independent Variable

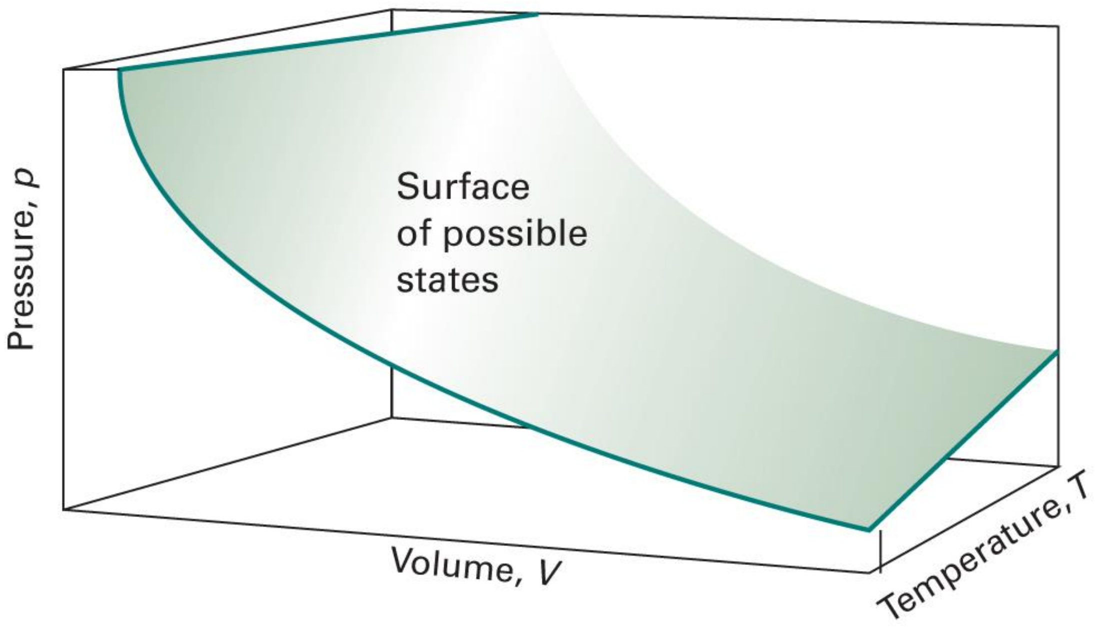
$$P = P(T, V, n) = \frac{nRT}{V}$$




(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 1A.5)

# Visualization: Two Independent Variables

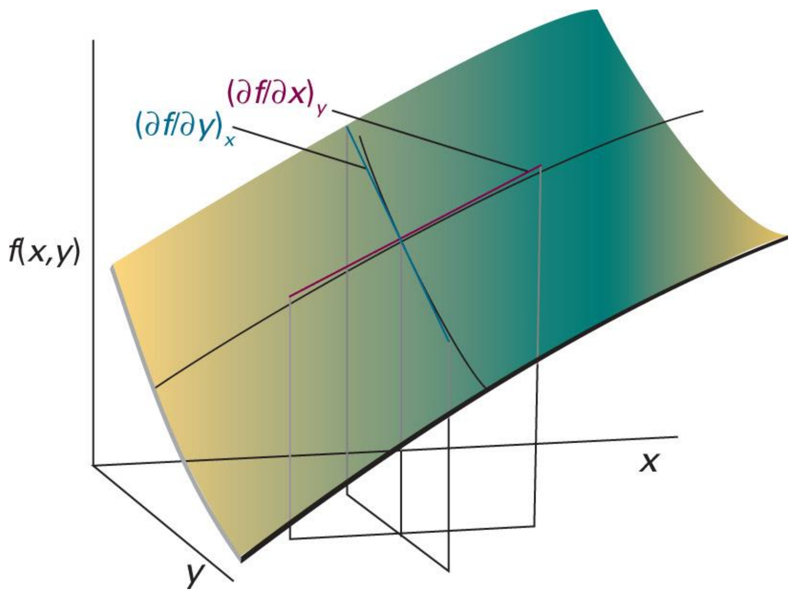
$$P = P(T, V, n) = \frac{nRT}{V}$$




(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 1A.6)

# Partial Derivatives (편미분)

"the slope of the function with respect to one of the variables, all the other variables being held constant"




$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x, y)}{\Delta y}$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 9, Sketch 1)

# Calculation of Partial Derivatives

Treat fixed variables like ordinary constants

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = \left(\frac{\partial}{\partial V}\left[\frac{nRT}{V}\right]\right)_{n,T} = -\frac{nRT}{V^2}$$

$$\left(\frac{\partial P}{\partial T}\right)_{n,V} = \left(\frac{\partial}{\partial T}\left[\frac{nRT}{V}\right]\right)_{n,V} = \frac{nR}{V}$$

$$\left(\frac{\partial P}{\partial n}\right)_{T,V} = \left(\frac{\partial}{\partial n}\left[\frac{nRT}{V}\right]\right)_{T,V} = \frac{RT}{V}$$

# Infinitesimal Changes

For $P = P(T, V, n)$,

$$\mathrm{d}P = \left(\frac{\partial P}{\partial V}\right)_{n,T} \mathrm{d}V \quad (n \text{ and } T \text{ fixed})$$

$$\mathrm{d}P = \left(\frac{\partial P}{\partial V}\right)_{n,T} \mathrm{d}V + \left(\frac{\partial P}{\partial T}\right)_{n,V} \mathrm{d}T \quad (n \text{ fixed})$$

$$\mathrm{d}P = \left(\frac{\partial P}{\partial V}\right)_{n,T} \mathrm{d}V + \left(\frac{\partial P}{\partial T}\right)_{n,V} \mathrm{d}T + \left(\frac{\partial P}{\partial n}\right)_{T,V} \mathrm{d}n$$

(total) differential of $P$ ($P$에 대한 전미분)

# Fundamental Equation of Differential Calculus

(미분학의 기본 방정식)

- $n$ independent variables: $x_1, x_2, \dots, x_n$
- dependent variable: $y$

The differential of $y$ is

$$\mathrm{d}y = \sum_{i=1}^{n} \left( \frac{\partial y}{\partial x_i} \right)_{x'} \mathrm{d}x_i$$

where $x'$ means keeping all of the variables except for $x_i$ fixed

# Case of Ideal Gas

$$\mathrm{d}P = \left(\frac{\partial P}{\partial V}\right)_{n,T} \mathrm{d}V + \left(\frac{\partial P}{\partial T}\right)_{n,V} \mathrm{d}T + \left(\frac{\partial P}{\partial n}\right)_{T,V} \mathrm{d}n$$

$$\mathrm{d}P = -\frac{nRT}{V^2}\mathrm{d}V + \frac{nR}{V}\mathrm{d}T + \frac{RT}{V}\mathrm{d}n$$

Approximately,

$$\Delta P \approx -\frac{nRT}{V^2}\Delta V + \frac{nR}{V}\Delta T + \frac{RT}{V}\Delta n$$

# Example 8.1

Calculate the approximate change in pressure of an ideal gas if

- the volume is changed from 20.000 L to 19.800 L,
- the temperature is changed from 298.15 K to 299.00 K, and
- the amount of gas in moles is changed from 1.0000 mol to 1.0015 mol.

Compare the result with the correct value of the pressure change.

$$\Delta P \approx -\frac{nRT}{V^2}\Delta V + \frac{nR}{V}\Delta T + \frac{RT}{V}\Delta n$$

In SI units, $R$ = 8.3145 J K$^{-1}$ mol$^{-1}$, and

$V$ = 0.020000 m$^3$, $T$ = 298.15 K, $n$ = 1.0000 mol (initial values);

$\Delta V$ = -2.00×10$^{-4}$ m$^3$, $\Delta T$ = 0.85 K, $\Delta n$ = 1.5×10$^{-3}$ mol

Substitution leads to $\Delta P$ = 1.779×10$^3$ N m$^{-2}$ = 1.779×10$^3$ Pa

# Example 8.1

Calculate the approximate change in pressure of an ideal gas if
- the volume is changed from 20.000 L to 19.800 L,
- the temperature is changed from 298.15 K to 299.00 K, and
- the amount of gas in moles is changed from 1.0000 mol to 1.0015 mol.

Compare the result with the correct value of the pressure change.

Approximate value: $\Delta P$ = 1.779×10$^3$ N m$^{-2}$ = 1.779×10$^3$ Pa

Correct value:

$$\Delta P = P_2 - P_1 = \frac{n_2 R T_2}{V_2} - \frac{n_1 R T_1}{V_1}$$

Substitution leads to $\Delta P$ = 1.797×10$^3$ Pa. The error is about 1%.

# In General

For $U = U(P, T, n)$,

$$\mathrm{d}U = \left(\frac{\partial U}{\partial P}\right)_{T,n} \mathrm{d}P + \left(\frac{\partial U}{\partial T}\right)_{P,n} \mathrm{d}T + \left(\frac{\partial U}{\partial n}\right)_{P,T} \mathrm{d}n$$

Approximately,

$$\Delta U \approx \left(\frac{\partial U}{\partial P}\right)_{T,n} \Delta P + \left(\frac{\partial U}{\partial T}\right)_{P,n} \Delta T + \left(\frac{\partial U}{\partial n}\right)_{P,T} \Delta n$$

# Example 8.2

For a sample of 1.000 mol of liquid $CCl_4$ at a temperature of 20°C and a pressure of 1.000 atm, the following experimental values are available:

$$(\partial U/\partial T)_{P,n} = 129.4\ \mathrm{J\ K^{-1}},\ (\partial U/\partial P)_{T,n} = 8.51\times10^{-4}\ \mathrm{J\ atm^{-1}}$$

Estimate the change in the energy of 1.000 mol of $CCl_4$ if its temperature is changed from 20.0 to 40.0°C and its pressure from 1.0 to 100.0 atm.

Since $n$ is fixed,

$$\Delta U \approx \left(\frac{\partial U}{\partial P}\right)_{T,n}\Delta P + \left(\frac{\partial U}{\partial T}\right)_{P,n}\Delta T + \left(\frac{\partial U}{\partial n}\right)_{P,T}\Delta n = \left(\frac{\partial U}{\partial P}\right)_{T,n}\Delta P + \left(\frac{\partial U}{\partial T}\right)_{P,n}\Delta T$$

From the given values, $\Delta T$ = 20.0 K and $\Delta P$ = 99.0 atm

Substitution leads to $\Delta U$ = 2588 J + 0.084 J = 2588 J

# Change of Variables

We can use different sets of independent variables:

- $U = U(T, V, n)$

$$\mathrm{d}U = \left(\frac{\partial U}{\partial T}\right)_{V,n} \mathrm{d}T + \left(\frac{\partial U}{\partial V}\right)_{T,n} \mathrm{d}V + \left(\frac{\partial U}{\partial n}\right)_{T,V} \mathrm{d}n$$

- $U = U(T, P, n)$

$$\mathrm{d}U = \left(\frac{\partial U}{\partial T}\right)_{P,n} \mathrm{d}T + \left(\frac{\partial U}{\partial P}\right)_{T,n} \mathrm{d}P + \left(\frac{\partial U}{\partial n}\right)_{T,P} \mathrm{d}n$$

# Be Careful

For most systems,

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} \neq \left(\frac{\partial U}{\partial T}\right)_{P,n}$$

Fixed variables (subscripts) are important!

# Example 8.3

Express the function $z = z(x, y) = ax^2 + bxy + cy^2$ in terms of $x$ and $u$, where $u = xy$. Find the two partial derivatives $(\partial z/\partial x)_y$ and $(\partial z/\partial x)_u$.

As $y = u/x$, substitution leads to a change of variables:

$$z(x,u) = ax^2 + bx\left(\frac{u}{x}\right) + c\left(\frac{u}{x}\right)^2 = ax^2 + bu + \frac{cu^2}{x^2}$$

Now, the partial derivatives are

$$\left(\frac{\partial z}{\partial x}\right)_y = 2ax + by$$

different!

$$\left(\frac{\partial z}{\partial x}\right)_u = 2ax - \frac{2cu^2}{x^3} = 2ax - \frac{2cy^2}{x}$$

# Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

Start from one set of independent variables:

$$U = U(T, P, n)$$

$$\mathrm{d}U = \boxed{\left(\frac{\partial U}{\partial T}\right)_{P,n}} \mathrm{d}T + \left(\frac{\partial U}{\partial P}\right)_{T,n} \mathrm{d}P + \left(\frac{\partial U}{\partial n}\right)_{T,P} \mathrm{d}n$$

# Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

$$\mathrm{d}U = \left(\frac{\partial U}{\partial T}\right)_{P,n} \mathrm{d}T + \left(\frac{\partial U}{\partial P}\right)_{T,n} \mathrm{d}P + \left(\frac{\partial U}{\partial n}\right)_{T,P} \mathrm{d}n$$

Divide both sides by $\mathrm{d}T$:

$$\frac{\mathrm{d}U}{\mathrm{d}T} = \left(\frac{\partial U}{\partial T}\right)_{P,n} \underbrace{\cancel{\frac{\mathrm{d}T}{\mathrm{d}T}}}_{1} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \frac{\mathrm{d}P}{\mathrm{d}T} + \left(\frac{\partial U}{\partial n}\right)_{T,P} \frac{\mathrm{d}n}{\mathrm{d}T}$$

# Relating Different Sets of Subscripts

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + ?$$

$$\frac{\mathrm{d}U}{\mathrm{d}T} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n}\frac{\mathrm{d}P}{\mathrm{d}T} + \left(\frac{\partial U}{\partial n}\right)_{T,P}\frac{\mathrm{d}n}{\mathrm{d}T}$$

Apply the constraints:

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n}\left(\frac{\partial P}{\partial T}\right)_{V,n} + \left(\frac{\partial U}{\partial n}\right)_{T,P}\underbrace{\cancel{\left(\frac{\partial n}{\partial T}\right)_{V,n}}}_{0}$$

# Variable-Change Identity

## (변수-변화 항등식)

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n}\left(\frac{\partial P}{\partial T}\right)_{V,n}$$

# Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z/\partial x)_u$ and $(\partial z/\partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + ?$$

For $z = z(x, y)$,

$$\mathrm{d}z = \left(\frac{\partial z}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial z}{\partial y}\right)_x \mathrm{d}y$$

$$\frac{\mathrm{d}z}{\mathrm{d}x} = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \frac{\mathrm{d}y}{\mathrm{d}x}$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

# Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z/\partial x)_u$ and $(\partial z/\partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

The function in Example 8.3 is $z = z(x, y) = ax^2 + bxy + cy^2$ and $u = xy$.

Hence,

$$\left(\frac{\partial z}{\partial y}\right)_x = bx + 2cy$$

$$\left(\frac{\partial y}{\partial x}\right)_u = \left(\frac{\partial}{\partial x}\left[\frac{u}{x}\right]\right)_u = -\frac{u}{x^2} = -\frac{y}{x}$$

# Example 8.4

Apply the foregoing method to the function in Example 8.3 and find the relation between $(\partial z/\partial x)_u$ and $(\partial z/\partial x)_y$.

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_u$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y + (bx + 2cy)\left(-\frac{y}{x}\right)$$

$$\left(\frac{\partial z}{\partial x}\right)_u = \left(\frac{\partial z}{\partial x}\right)_y - by - \frac{2cy^2}{x}$$

# Reciprocal Identity

## (역 항등식)

$$\left(\frac{\partial y}{\partial x}\right)_{z,u} = \frac{1}{(\partial x/\partial y)_{z,u}}$$

# Example 8.5

Show that

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = \frac{1}{(\partial V/\partial P)_{n,T}}$$

for an ideal gas.

Since $P = nRT/V$,

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = -\frac{nRT}{V^2}$$

Since $V = nRT/P$,

$$\left(\frac{\partial V}{\partial P}\right)_{n,T} = -\frac{nRT}{P^2} = -nRT\left(\frac{V}{nRT}\right)^2 = -\frac{V^2}{nRT}$$

# Second Partial Derivatives

## (2계 편도함수)

For $z = z(x, y)$,

$$\left(\frac{\partial^2 z}{\partial x^2}\right)_y = \left[\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)_y\right]_y$$

$$\frac{\partial^2 z}{\partial y \partial x} = \left[\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)_y\right]_x$$

(Mixed second partial derivative; 혼합 2계 편도함수)

# Euler Reciprocity Relation

## (오일러 역관계식)

For $z = z(x, y)$,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

# Example 8.6

Show that

$$\left(\frac{\partial^2 P}{\partial V \partial T}\right)_n = \left(\frac{\partial^2 P}{\partial T \partial V}\right)_n$$

for an ideal gas.

Since $P = nRT/V$,

$$\left(\frac{\partial^2 P}{\partial V \partial T}\right)_n = \left[\frac{\partial}{\partial V}\left(\frac{nR}{V}\right)\right]_{n,T} = -\frac{nR}{V^2}$$

and

$$\left(\frac{\partial^2 P}{\partial T \partial V}\right)_n = \left[\frac{\partial}{\partial T}\left(-\frac{nRT}{V^2}\right)\right]_{n,V} = -\frac{nR}{V^2}$$

# Thermodynamic Relation

$$\mathrm{d}U = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}n$$

For $U = U(S, V, n)$,

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_{V,n} \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_{S,n} \mathrm{d}V + \left(\frac{\partial U}{\partial n}\right)_{S,V} \mathrm{d}n$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,n}, \qquad P = -\left(\frac{\partial U}{\partial V}\right)_{S,n}, \qquad \mu = \left(\frac{\partial U}{\partial n}\right)_{S,V}$$

# Maxwell Relations

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,n}, \qquad P = -\left(\frac{\partial U}{\partial V}\right)_{S,n}, \qquad \mu = \left(\frac{\partial U}{\partial n}\right)_{S,V}$$

Since $(\partial^2 U/\partial V \partial S)_n = (\partial^2 U/\partial S \partial V)_n$,

$$\left(\frac{\partial T}{\partial V}\right)_{S,n} = -\left(\frac{\partial P}{\partial S}\right)_{V,n}$$

Relation between apparently unrelated quantities

# Cycle Rule

## (순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

Start from $y = y(x, z)$;

$$\mathrm{d}y = \left(\frac{\partial y}{\partial x}\right)_z \mathrm{d}x + \left(\frac{\partial y}{\partial z}\right)_x \mathrm{d}z$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{\partial y}{\partial x}\right)_z \underbrace{\cancel{\frac{\mathrm{d}x}{\mathrm{d}x}}}_{1} + \left(\frac{\partial y}{\partial z}\right)_x \frac{\mathrm{d}z}{\mathrm{d}x}$$

# Cycle Rule

## (순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \frac{\mathrm{d}z}{\mathrm{d}x}$$

Fix $y$;

$$\underbrace{\cancel{\left(\frac{\partial y}{\partial x}\right)_y}}_{0} = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y$$

# Cycle Rule

## (순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

$$0 = \left(\frac{\partial y}{\partial x}\right)_z + \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y$$

$$-1 = \frac{\left(\frac{\partial y}{\partial x}\right)_z}{\left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y} = \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x$$

# Example 8.8

For the function $y = z \ln x$, show that the cycle rule is valid.

Since $x = e^{y/z}$ and $z = y/(\ln x)$,

$$\left(\frac{\partial y}{\partial x}\right)_z = \frac{z}{x}, \qquad \left(\frac{\partial x}{\partial z}\right)_y = -\frac{y}{z^2} e^{y/z}, \qquad \left(\frac{\partial z}{\partial y}\right)_x = \frac{1}{\ln x}$$

Hence,

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = \frac{z}{x} \times \left(-\frac{y}{z^2} e^{y/z}\right) \times \frac{1}{\ln x}$$

$$= \frac{z}{x} \times \left(-\frac{y}{z^2} x\right) \times \frac{z}{y} = -1$$

# Chain Rule

## (연쇄 법칙)

For $z = z(u, x, y)$ and $x = x(u, v, y)$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Note:

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} \left(\frac{\partial y}{\partial z}\right)_{u,v} = 1$$

Check the subscripts!

# Example 8.9

Show that if $z = ax^2 + bvx$ and $x = uy$, then the chain rule is valid.

Recall the chain rule:

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Since $z = z(y, u, v) = a(uy)^2 + bv(uy) = au^2y^2 + buvy$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = 2au^2y + buv$$

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} = 2ax + bv, \qquad \left(\frac{\partial x}{\partial y}\right)_{u,v} = u$$

same!

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} = (2ax + bv) \times u = 2au^2y + buv$$